UNIT 7 · TOPIC 11

Energy Requirement in Agriculture — Energy Ratio and Specific Energy

Syllabus topic: Energy requirement in agricultural production systems; energy ratio and specific energy value

Forms and classification of energy inputs

Classification	Classes
DIRECT vs INDIRECT	Direct — used in operations: diesel/petrol, electricity, human labour, draft-animal power. Indirect (embodied) — used to MANUFACTURE inputs: fertilisers (the LARGEST indirect input, especially nitrogen at ~60–78 MJ per kg N), seeds, agro-chemicals, machinery
COMMERCIAL vs NON-COMMERCIAL	Commercial: diesel, electricity, fertiliser. Non-commercial: human, animal, firewood, dung
RENEWABLE vs NON-RENEWABLE	Renewable: human, animal, biomass, solar. Non-renewable: fossil fuels, most fertiliser and electricity

Common unit — the megajoule

Input	Energy equivalent
Diesel (incl. embodied)	~56 MJ/L
Human labour	~1.9 MJ/h
Bullock pair	~10 MJ/h
Electricity (incl. generation)	~11.9 MJ/kWh
Nitrogen fertiliser	~60–78 MJ/kg N

Energy indices

ENERGY RATIO (output–input ratio, energy-use efficiency) = energy output (in produce) ÷ energy input

SPECIFIC ENERGY = energy input ÷ yield (MJ/kg) — lower is better

ENERGY PRODUCTIVITY = yield ÷ energy input (kg/MJ); ENERGY INTENSITY = input per hectare (MJ/ha); NET ENERGY = output − input

- Energy ratio > 1 means a net energy gain. Traditional systems are often **2–5**; intensive irrigated systems can fall toward or below 1.

Worked example — wheat.
1. Inputs = 18,000 MJ/ha; grain yield 4 t/ha at ~14.7 MJ/kg → output = 58,800 MJ/ha (plus straw).
2. Energy ratio (grain only) = 58,800/18,000 = **3.3**.
3. Specific energy = 18,000/4000 = **4.5 MJ/kg**.
4. Energy productivity = 1/4.5 = **0.22 kg/MJ**.

Practice MCQs

Q129. DIRECT energy in agriculture includes:
(a) Fertiliser manufacture
(b) Diesel, electricity, human and animal power
(c) Seed embodied energy
(d) Machinery manufacture

Q130. The LARGEST indirect (embodied) energy input in most crops is:
(a) Seed
(b) Nitrogen fertiliser
(c) Pesticide
(d) Machinery

Q131. Human, animal, biomass and solar energy are classified as:
(a) Non-renewable
(b) Renewable
(c) Commercial only
(d) Fossil

Q132. Diesel, electricity and fertiliser are examples of:
(a) Non-commercial energy
(b) Commercial energy
(c) Renewable energy
(d) Muscle power

Q133. Energy ratio (energy-use efficiency) is:
(a) Input ÷ output
(b) Output ÷ input
(c) Output − input
(d) Output × input

Q134. An energy ratio greater than 1 means:
(a) Net energy loss
(b) Net energy gain
(c) No output
(d) Zero input

Q135. Specific energy is defined as:
(a) Input ÷ yield (MJ/kg)
(b) Yield ÷ input
(c) Output ÷ area
(d) Input × yield

Q136. Inputs 18,000 MJ/ha and grain energy output 58,800 MJ/ha give an energy ratio of about:
(a) 1.3
(b) 2.0
(c) 3.3
(d) 5.0

Q137. For the same crop (4 t/ha, 18,000 MJ/ha), specific energy is about:
(a) 1.5
(b) 4.5 MJ/kg
(c) 14.7
(d) 45

Q138. Energy productivity is the reciprocal of specific energy, in units of:
(a) MJ/kg
(b) kg/MJ
(c) MJ/ha
(d) kg/ha

Answers
Q129 — (b) Diesel, electricity, human and animal power
Q130 — (b) Nitrogen fertiliser
Q131 — (b) Renewable
Q132 — (b) Commercial energy
Q133 — (b) Output ÷ input

Q134 — (b) Net energy gain

Q135 — (a) Input ÷ yield (MJ/kg)

Q136 — (c) 3.3 — *Energy ratio = 58,800/18,000 = 3.3.*

Q137 — (b) 4.5 MJ/kg — *Specific energy = 18,000/4000 = 4.5 MJ/kg.*

Q138 — (b) kg/MJ — *Productivity = 1/4.5 = 0.22 kg/MJ.*

Sources: Rai — Non-Conventional Energy Sources; Mittal — Energy in Agriculture; MNRE Biogas/SATAT programmes & PIB (as given in the source notes, July 2026); IS 16087:2016 (CBG specification); standard bioenergy texts; NPTEL bioenergy & energy-management lectures.